Surname ______________________	**Initials** ______________________
Van	**Voorletters**
	Stud. No/*nr*. ______________________
The undersigned hereby confirms that they are familiar with and accepts the examination regulations of the University of Pretoria	*Ondergetekende bevestig hiermee dat hulle bewus is van die eksamenregulasies van die Universiteit van Pretoria en dat hy/sy hom/haar daaraan onderwerp.*

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Class Test 3 **20 May / *Mei* 2013** ***Klastoets 3***

FSK116

Total/Totaal: 30

Question 1/*Vraag 1*

A 4.0 kg object slides along a frictionless surface and explodes into two 2.0 kg parts: one moves at 3.0 m/s due north, and the other at 6.0 m/s, 30° North of East. Determine the original speed of the object.

'n 4.0 kg voorwerp gly oor 'n wrywinglose oppervlak en ontplof in twee 2.0 kg dele: die een beweeg teen 3.0 m / s Noord, en die ander een beweeg teen 6.0 m / s, 30° Noord van Oos. Bepaal die oorspronklike spoed van die voorwerp. [10]

Our $+x$ direction is east and $+y$ direction is north.
The linear momenta for the two m = 2.0 kg parts are then

$$\vec{p}_1 = m\vec{v}_1 = mv_1\,\hat{\mathrm{j}} = 2.0\text{ kg }(3.0\text{ m/s}) = 6.0\text{ kg.m/s} \quad ✓$$

and

$$\vec{p}_2 = m\vec{v}_2 = m\left(v_{2x}\,\hat{\mathrm{i}} + v_{2y}\,\hat{\mathrm{j}}\right) = mv_2\left(\cos\theta\,\hat{\mathrm{i}} + \sin\theta\,\hat{\mathrm{j}}\right)$$

$$\vec{p}_2 = \left((2.0\text{ kg})(6.0\text{ m/s})(\cos 30°)\right)\hat{\mathrm{i}} + \left((2.0\text{ kg})(6.0\text{ m/s})(\sin 30°)\right)\hat{\mathrm{j}} \quad ✓✓$$

$$= \left(10.4\,\text{kg.m.s}^{-1}\right)\hat{\mathrm{i}} + \left(6\,\text{kg.m.s}^{-1}\right)\hat{\mathrm{j}} \quad ✓✓$$

The combined linear momentum of both parts is then

$$\vec{P} = \vec{p}_1 + \vec{p}_2 = \left(6\,\text{kg.m.s}^{-1}\right)\hat{\mathrm{j}} + \left(\left(10.4\,\text{kg.m.s}^{-1}\right)\hat{\mathrm{i}} + \left(6\,\text{kg.m.s}^{-1}\right)\hat{\mathrm{j}}\right) \quad ✓$$

$$= \left(10\hat{\mathrm{i}} + 12\hat{\mathrm{j}}\right)\text{ kg}\cdot\text{m/s}. \quad ✓✓$$

From conservation of linear momentum we know that this is also the linear momentum of the whole sistem before it splits. Thus the speed of the 4.0-kg sistem is

$$v = \frac{P}{M} = \frac{\sqrt{P_x^2 + P_y^2}}{M} = \frac{\sqrt{(10\text{ kg}\cdot\text{m/s})^2 + (12\text{ kg}\cdot\text{m/s})^2}}{4.0\text{ kg}} = 3.9\text{ m/s.} \quad ✓✓$$

Question 2/*Vraag 2*

Starting from rest at $t = 0$ s, a wheel undergoes a constant angular acceleration. At $t = 2.0$s, the angular velocity of the wheel is 5.0 rad/s. The acceleration continues until $t = 20$ s, when it abruptly ceases. Determine the angular acceleration during the first 20 s and the angle through which the wheel rotates during the interval $t = 0$ s to $t = 40$ s?

'n Wiel aanvanklik in rus by t = 0 s ondergaan konstante hoek-versnelling. By 2.0 s het die wiel 'n hoek-snelheid van 5.0 rad/s. Die versnelling gaan voort tot by t = 20 s, wanneer dit skielik ophou. Bepaal die versnelling tydens die eerste 20 s en die hoek waardeur die wiel roteer gedurende die tydsinterval t = 0 s tot t = 40 s. [6]

$\omega_0 = 0$ rad/s $\quad\quad \omega_2 = 5.0$ rad/s $\quad\quad t = 2.0$ s

$$\alpha = \frac{\omega - \omega_0}{t - t_0} = \frac{5\,\text{rad/s}}{2\,\text{s} - 0\,\text{s}} = 2.5\,\text{rad/s}^2$$ ✓✓

This acceleration lasts for 20 s only.

During $t = 0$ s to $t = 20$ s There is the following angular displacement:

$$\begin{aligned}\Delta\theta_{20} &= \omega_0 t + \tfrac{1}{2}\alpha t^2 \\ &= 0 + \tfrac{1}{2}\left(2.5\,\text{rad/s}^2\right)\left(20\,\text{s}\right)^2 \\ &= 25\,\text{rad}\end{aligned}$$ ✓

During $t = 20$ s to $t = 40$ s there is no acceleration but there is constant angular velocity:

$$\begin{aligned}\omega_{20} &= \omega_0 + \alpha t \\ &= 0 + \left(2.5\,\text{rad/s}\right)\left(20\,\text{s}\right) \\ &= 50\,\text{rad/s}\end{aligned}$$

This is the angular velocity after 20 s ✓

$\omega_{20} = 50$ rad/s $\quad\quad \alpha = 0.0$ rad/s $\quad\quad \Delta t = 20.0$ s

At this constant angular velocity – we get the following angular dislplacement for the next 20 s.

$$\begin{aligned}\Delta\theta_2 &= \omega_{20} t = \left(50\,\text{rad/s}\right)\left(20\,\text{s}\right) \\ &= 1\times10^3\,\text{rad}\end{aligned}$$ ✓

Total angle:

$$\begin{aligned}\theta &= \theta_1 + \theta_2 \\ &= 25\,\text{rad} + 1000\,\text{rad} \\ &= 1025\,\text{rad}\end{aligned}$$

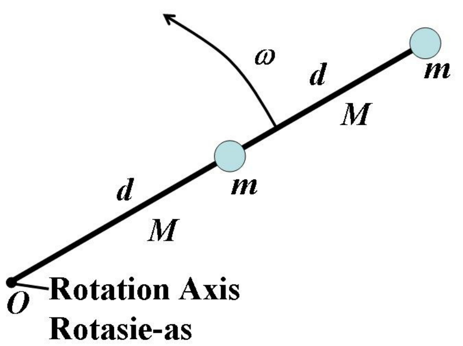
Remember to add the two angular displacements together ✓

to get the total angular displacement.

Question 3/*Vraag 3*

The figure shows two particles, each with mass $m = 0.85$ kg, which are fastened to each other and to a rotation axis at O by two thin rods, each with length $d = 5.6$ cm and mass $M = 1.2$ kg. The combination rotates around the rotation axis with the angular speed $\omega = 0.30$ rad/s.

Die figuur dui twee deeltjies aan, elk met massa m = 0.85 kg, wat vas is aan mekaar en aan die rotasie-as by O deur middel van twee dun stafies, elk met lengte d = 5.6 cm en massa M = 1.2 kg. Die stelsel roteer om die rotasie-as met 'n hoek-snelheid ω = 0.30 rad/s.



a) Determine the system's rotational inertia about O?
Bepaal die sisteem se traagheidsmoment om O? [7]

Consider particles as point masses:	
Particle1: $I_1 = mR_1^2$	✓
Particle2: $I_2 = mR_2^2$	✓
Rod 1: $I_{R1} = \frac{1}{12}ML^2 + Mh_1^2$	✓
Rod 2: $I_{R2} = \frac{1}{12}ML^2 + Mh_1^2$	✓
Total rotational inertia:	
$I = I_1 + I_2 + I_{R1} + I_{R2}$	
$= m\left(R_1^2 + R_2^2\right) + \left(\frac{1}{12}ML^2 + Mh_1^2\right) + \left(\frac{1}{12}ML^2 + Mh_2^2\right)$	✓
$= m\left(R_1^2 + R_2^2\right) + 2\left(\frac{1}{12}ML^2\right) + M\left(h_1^2 + h_2^2\right)$	
$= (0.85\,\text{kg})\left(\left(5.6\times10^{-2}\,\text{m}\right)^2 + \left(11.2\times10^{-2}\,\text{m}\right)^2\right) + 2\left(\frac{1}{12}(1.2\,\text{kg})\left(5.6\times10^{-2}\,\text{m}\right)^2\right)$	✓
$+ (1.2\,\text{kg})\left(\left(5.6\times10^{-2}\,\text{m}\right)^2 + \left(8.4\times10^{-2}\,\text{m}\right)^2\right)$	
$= 0.023\,\text{kg}\cdot\text{m}^3$	✓

b) Determine the system's kinetic energy.
Bepaal die stelsel se kinetiese energie. [3]

$K = ½I\omega^2 = ½\left(0.023\,\text{kg}\cdot\text{m}^3\right)\left(0.30\,\text{rad/s}\right)^2$	✓
$= 1.1\times10^{-3}\ \text{J}$	✓✓
Use the value of I from part (a)	

c) Determine the constant torque necessary to bring the system to rest in 2.0 s
Bepaal die konstante draaimoment nodig om die stelsel tot rus te bring in 2.0 s. [4]

$\omega_0 = 0.30$ rad/s $\quad \omega = 0$ rad/s $\quad t = 2.0$ s	
$\theta = ½\left(\omega + \omega_0\right)t = ½\left(0 + 0.3\,\text{rad/s}\right)2.0\,\text{s}$	
$= 0.3\,\text{rad}$	✓
$W = \tau\theta = \Delta K$	✓
$\tau = \frac{\Delta K}{\theta} = \frac{1.1\times10^{-3}\ \text{J}}{0.3\,\text{rad}}$	
$= 0.0037\,\text{N}\cdot\text{m}$	✓✓
Can also use the following method:	

$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{0.30\,\text{rad/s}}{2.0\,\text{s}} = 0.15\,\text{rad/s}^2$ $\tau = I\alpha = 0.023\,\text{kg}\cdot\text{m}^3\left(0.15\,\text{rad/s}^2\right)$ $= 0.0035\,\text{N}\cdot\text{m}$ Slightly different value due to rounding in (a)	

That's it for today / Dit is al vir vandag